

The variance of the trade schedule is:

$$\mathfrak{R}^2(r_k) = \sum_{j=1}^n R_j^2 \cdot t^* \cdot \frac{\sigma^2}{n} \cdot P_0^2 \quad (7.38)$$

By Equations 7.36 and 7.38 we have:

$$\mathfrak{R}^2(r_k) = \sum_{j=1}^n X^2 \left(1 - \frac{(j-1)}{n}\right)^2 \cdot t^* \cdot \frac{\sigma^2}{n} \cdot P_0^2 \quad (7.39)$$

By factoring we have:

$$\mathfrak{R}^2(r_k) = X^2 \cdot P_0^2 \cdot t^* \cdot \frac{\sigma^2}{n} \cdot \sum_{j=1}^n \left(1 - \frac{(j-1)}{n}\right)^2 \quad (7.40)$$

And by expansion we have:

$$\mathfrak{R}^2(r_k) = X^2 \cdot P_0^2 \cdot t^* \cdot \frac{\sigma^2}{n} \cdot \sum_{j=1}^n \left(1 - \frac{2(j-1)}{n} + \frac{(j-1)^2}{n^2}\right) \quad (7.41)$$

Using the following identities:

$$\begin{aligned} \sum_{j=1}^n 1 &= n \\ \sum_{j=1}^n x &= \frac{n(n+1)}{2} \\ \sum_{j=1}^n x^2 &= \frac{n(n+1)(2n+1)}{6} \end{aligned}$$

Equation 7.41 is now:

$$\mathfrak{R}^2 = X^2 \cdot P_0^2 \cdot t^* \cdot \sigma^2 \cdot \frac{1}{n} \left(n - (n-1) + \frac{(n-1)(2n-1)}{6n^2} \right) \quad (7.42)$$

This further reduces to:

$$\mathfrak{R}^2 = X^2 \cdot P_0^2 \cdot t^* \cdot \sigma^2 \cdot \left(\frac{1}{3} + \frac{1}{n} + \frac{1}{2n} + \frac{1}{6n^2} \right) \quad (7.43)$$

Now if we let the number of trading periods over the defined trading time increase, the size of the trading interval becomes infinitely small and our trade schedule strategy approaches a continuous trade rate strategy.

To show this we take the limit as n approaches infinity in Equation 7.43:

$$\lim_{n \rightarrow \infty} X^2 \cdot P_0^2 \cdot t^* \cdot \sigma^2 \cdot \left(\frac{1}{3} + \frac{1}{n} + \frac{1}{2n} + \frac{1}{6n^2} \right) = X^2 \cdot t^* \cdot \sigma^2 \cdot \frac{1}{3} \quad (7.44)$$